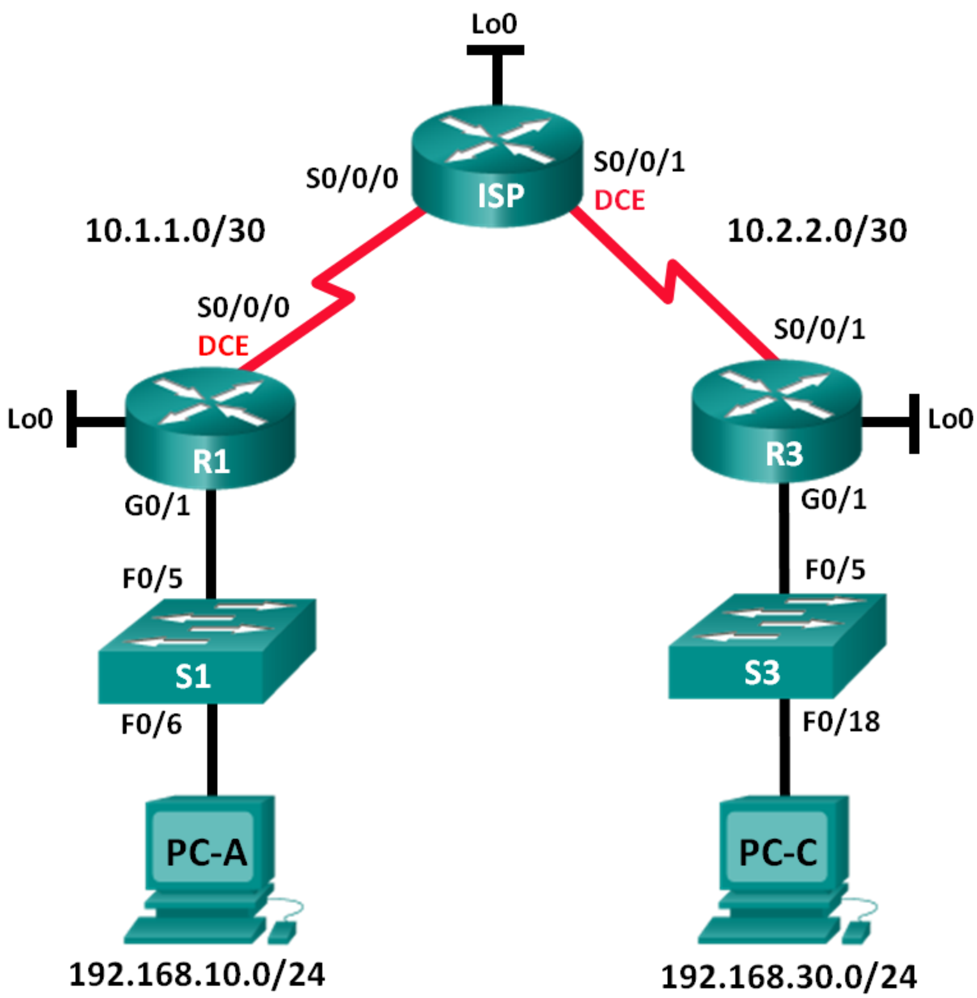
Lab 5 – Configuring and Verifying Standard IPv4 ACLs (5%)

Student Name: ____________GROUP G____________10 Marks

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/1	192.168.10.1	255.255.255.0	N/A
	Lo0	192.168.20.1	255.255.255.0	N/A
	S0/0/0 (DCE)	10.1.1.1	255.255.255.252	N/A
ISP	S0/0/0	10.1.1.2	255.255.255.252	N/A
	S0/0/1 (DCE)	10.2.2.2	255.255.255.252	N/A
	Lo0	209.165.200.225	255.255.255.224	N/A
R3	G0/1	192.168.30.1	255.255.255.0	N/A
	Lo0	192.168.40.1	255.255.255.0	N/A
	S0/0/1	10.2.2.1	255.255.255.252	N/A
S1	VLAN 1	192.168.10.11	255.255.255.0	192.168.10.1
S3	VLAN 1	192.168.30.11	255.255.255.0	192.168.30.1
PC-A	NIC	192.168.10.3	255.255.255.0	192.168.10.1
PC-C	NIC	192.168.30.3	255.255.255.0	192.168.30.1

Objectives

Part 1: Set Up the Topology and Initialize Devices

- Set up equipment to match the network topology.
- Initialize and reload the routers and switches.

Part 2: Configure Devices and Verify Connectivity

- Assign a static IP address to PCs.
- Configure basic settings on routers.
- Configure basic settings on switches.
- Configure OSPF routing on R1, ISP, and R3.
- Verify connectivity between devices.

Part 3: Configure and Verify Standard Numbered and Named ACLs

- Configure, apply, and verify a numbered standard ACL.
- Configure, apply, and verify a named ACL.

Part 4: Modify a Standard ACL

- Modify and verify a named standard ACL.
- Test the ACL.

Background / Scenario

Network security is an important issue when designing and managing IP networks. The ability to configure proper rules to filter packets, based on established security policies, is a valuable skill.

In this lab, you will set up filtering rules for two offices represented by R1 and R3. Management has established some access policies between the LANs located at R1 and R3, which you must implement. The ISP router sitting between R1 and R3 will not have any ACLs placed on it. You would not be allowed any administrative access to an ISP router because you can only control and manage your own equipment.

Note: The routers used with CCNA hands-on labs are Cisco 1941 Integrated Services Routers (ISRs) with Cisco IOS Release 15.2(4)M3 (universalk9 image). The switches used are Cisco Catalyst 2960s with Cisco IOS Release 15.0(2) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

Note: Make sure that the routers and switches have been erased and have no startup configurations. If you are unsure, contact your instructor.

Part 1: Set Up the Topology and Initialize Devices

In Part 1, you set up the network topology and clear any configurations, if necessary.

Part 2: Configure Devices and Verify Connectivity

In Part 2, you configure basic settings on the routers, switches, and PCs. Refer to the Topology and Addressing Table for device names and address information.

Step 1: Configure IP addresses on PC-A and PC-C.

Step 2: Configure basic settings for the routers.

a. Console into the router and enter global configuration mode.

b. Copy the following basic configuration and paste it to the running-configuration on the router.

```
no ip domain-lookup
hostname R1
service password-encryption
enable secret class
banner motd #
Unauthorized access is strictly prohibited. #
Line con 0
password class
login
logging synchronous
line vty 0 4
password class
login
```

c. Configure the device name as shown in the topology.

d. Create loopback interfaces on each router as shown in the Addressing Table.

e. Configure interface IP addresses as shown in the Topology and Addressing Table.

f. Assign a clock rate of **128000** to the DCE serial interfaces.

g. Enable Telnet access.

h. Copy the running configuration to the startup configuration.

Step 3: Configure basic settings on the switches.

a. Console into the switch and enter global configuration mode.

b. Copy the following basic configuration and paste it to the running-configuration on the switch.

```
no ip domain-lookup
service password-encryption
enable secret class
banner motd #
Unauthorized access is strictly prohibited. #
Line con 0
password cisco
login
logging synchronous
line vty 0 15
password cisco
login
exit
```

c. Configure the device name as shown in the topology.

d. Configure the management interface IP address as shown in the Topology and Addressing Table.

e. Configure a default gateway.

f. Enable Telnet access.

g. Copy the running configuration to the startup configuration.

Step 4: Configure Rip routing on R1, ISP, and R3.

a. Configure RIP version 2 and advertise all networks on R1, ISP, and R3. The OSPF configuration for R1 and ISP is included for reference.

```
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# network 192.168.10.0
R1(config-router)# network 192.168.20.0
R1(config-router)# network 10.1.1.0

ISP(config)# router rip
ISP(config-router)# version 2
ISP(config-router)# network 209.165.200.224
ISP(config-router)# network 10.1.1.0
ISP(config-router)# network 10.2.2.0
```

b. After configuring Rip on R1, ISP, and R3, verify that all routers have complete routing tables, listing all networks. Troubleshoot if this is not the case. **Show the routing tables here. [1 mark]**

COM5 - PuTTY

```
R1(config)#exit
R1#sh
*Jul 28 13:04:56.123: %SYS-5-CONFIG_I: Configured from console by consoleow
% Type "show ?" for a list of subcommands
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.10.0/24 is directly connected, GigabitEthernet0/1
L        192.168.10.1/32 is directly connected, GigabitEthernet0/1
      192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.20.0/24 is directly connected, Loopback0
L        192.168.20.1/32 is directly connected, Loopback0
R1#
```

COM3 - PuTTY

```
R3(config-if)#ip address 192.168.30.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#
*Jul 28 13:24:59.239: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed stat
e to down
*Jul 28 13:25:03.823: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed stat
e to up
*Jul 28 13:25:04.823: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEth
ernet0/1, changed state to up
R3(config-if)#interface Lo0
R3(config-if)#ip address
*Jul 28 13:27:56.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0,
 changed state to up
% Incomplete command.

R3(config-if)#ip address 192.168.40.1 255.255.255.0
R3(config-if)#no shutdown
R3(config-if)#interface S0/0/1
R3(config-if)#ip address 10.2.2.1 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#
*Jul 28 13:29:13.219: %LINK-3-UPDOWN: Interface Serial0/0/1, changed state to do
wn
R3(config-if)#exit
R3(config)#interface S0/0/0
R3(config-if)#clock rate 128000
R3(config-if)#exit
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#network 192.168.30.0
R3(config-router)#network 192.168.40.0
R3(config-router)#network 10
% Incomplete command.

R3(config-router)#network 10.2.2.0
R3(config-router)#exit
R3(config)#show ip route
             ^
% Invalid input detected at '^' marker.

R3(config)#exit
R3#ip
*Jul 28 13:40:25.099: %SYS-5-CONFIG_I: Configured from console by conso
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.30.0/24 is directly connected, GigabitEthernet0/1
L        192.168.30.1/32 is directly connected, GigabitEthernet0/1
      192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.40.0/24 is directly connected, Loopback0
L        192.168.40.1/32 is directly connected, Loopback0
R3#
```


Step 5: Verify connectivity between devices.

Note: It is very important to test whether connectivity is working **before** you configure and apply access lists! You want to ensure that your network is properly functioning before you start to filter traffic.

a. From PC-A, ping PC-C and the loopback interface on R3. Were your pings successful? __**Yes**_

b. From R1, ping PC-C and the loopback interface on R3. Were your pings successful? __**Yes**___

c. From PC-C, ping PC-A and the loopback interface on R1. Were your pings successful? _**Yes**_

d. From R3, ping PC-A and the loopback interface on R1. Were your pings successful? __**Yes**__

Insert here screen shots of successful or unsuccessful ping commands [1 mark]

Administrator: Command Prompt

```
C:\Users\ngharti>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:
Reply from 192.168.30.3: bytes=32 time=19ms TTL=125
Reply from 192.168.30.3: bytes=32 time=18ms TTL=125
Reply from 192.168.30.3: bytes=32 time=19ms TTL=125
Reply from 192.168.30.3: bytes=32 time=19ms TTL=125

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 19ms, Average = 18ms

C:\Users\ngharti>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:
Reply from 192.168.40.1: bytes=32 time=18ms TTL=253
Reply from 192.168.40.1: bytes=32 time=18ms TTL=253
Reply from 192.168.40.1: bytes=32 time=18ms TTL=253
Reply from 192.168.40.1: bytes=32 time=18ms TTL=253

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 18ms, Average = 18ms

C:\Users\ngharti>
```

COM5 - PuTTY

```
ocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down

GigabitEthernet0/0         unassigned      YES unset  administratively down down

GigabitEthernet0/1         192.168.10.1    YES manual up                    up

Serial0/0/0                10.1.1.1        YES manual up                    up

Serial0/0/1                unassigned      YES unset  administratively down down

Loopback0                  192.168.20.1    YES manual up                    up

R1#ping 192.168.30.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.30.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/32 ms
R1#ping 192.168.40.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.40.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/32 ms
R1#
```


COM3 - PuTTY

```
ocol
Embedded-Service-Engine0/0 unassigned      YES unset  administratively down down

GigabitEthernet0/0         unassigned      YES unset  administratively down down

GigabitEthernet0/1         192.168.30.1    YES manual up                    up

Serial0/0/0                unassigned      YES unset  administratively down down

Serial0/0/1                10.2.2.1        YES manual up                    up

Loopback0                  192.168.40.1    YES manual up                    up

R3>ping 192.168.10.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/28 ms
R3>ping 192.168.20.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/28/28 ms
R3>
```

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.19044.2486]
(c) Microsoft Corporation. All rights reserved.

C:\Users\GBASNET>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=18ms TTL=253
Reply from 192.168.10.1: bytes=32 time=18ms TTL=253
Reply from 192.168.10.1: bytes=32 time=18ms TTL=253
Reply from 192.168.10.1: bytes=32 time=18ms TTL=253

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 18ms, Average = 18ms

C:\Users\GBASNET>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:
Reply from 192.168.20.1: bytes=32 time=18ms TTL=253
Reply from 192.168.20.1: bytes=32 time=18ms TTL=253
Reply from 192.168.20.1: bytes=32 time=18ms TTL=253
Reply from 192.168.20.1: bytes=32 time=18ms TTL=253

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 18ms, Average = 18ms

C:\Users\GBASNET>
```

Part 3: Configure and Verify Standard Numbered and Named ACLs

Step 1: Configure a numbered standard ACL.

Standard ACLs filter traffic based on the source IP address only. A typical best practice for standard ACLs is to configure and apply it as close to the destination as possible. For the first access list, create a standard numbered ACL that allows traffic from all hosts on the 192.168.10.0/24 network and all hosts on the 192.168.20.0/24 network to access all hosts on the 192.168.30.0/24 network. The security policy also states that a **deny any** access control entry (ACE), also referred to as an ACL statement, should be present at the end of all ACLs.

What wildcard mask would you use to allow all hosts on the 192.168.10.0/24 network to access the 192.168.30.0/24 network?

________________**0.0.0.255**____________________________

Following Cisco's recommended best practices, on which router would you place this ACL? ___**R3**_____

On which interface would you place this ACL? In what direction would you apply it?

____

a. Configure the ACL on R3. Use 1 for the access list number.

```
R3(config)# access-list 1 remark Allow R1 LANs Access
R3(config)# access-list 1 permit 192.168.10.0 0.0.0.255
R3(config)# access-list 1 permit 192.168.20.0 0.0.0.255
R3(config)# access-list 1 deny any
```

b. Apply the ACL to the appropriate interface in the proper direction.

```
R3(config)# interface g0/1
R3(config-if)# ip access-group 1 out
```

c. Verify a numbered ACL.

The use of various **show** commands can aid you in verifying both the syntax and placement of your ACLs in your router.

To see access list 1 in its entirety with all ACEs, which command would you use?

______________**show access-list 1**________________________

What command would you use to see where the access list was applied and in what direction?

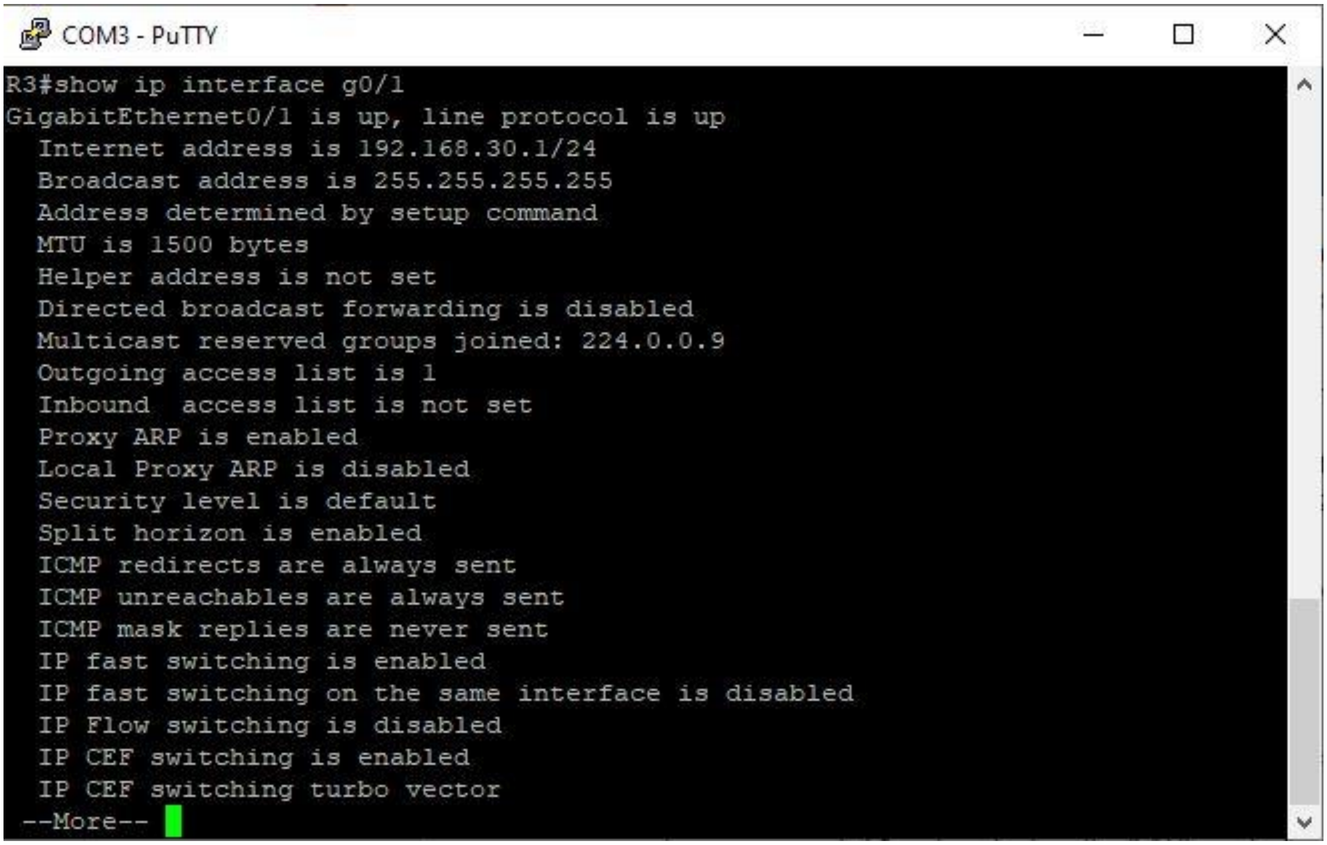
______**show ip interface**______________________

1) On R3, issue the **show access-lists 1** command.

```
R3# show access-list 1
Standard IP access list 1
    10 permit 192.168.10.0, wildcard bits 0.0.0.255
    20 permit 192.168.20.0, wildcard bits 0.0.0.255
    30 deny   any
```

2) On R3, issue the **show ip interface g0/1** command. Verify that the access list 1 is included

Insert here screen shots of the command [1 mark]

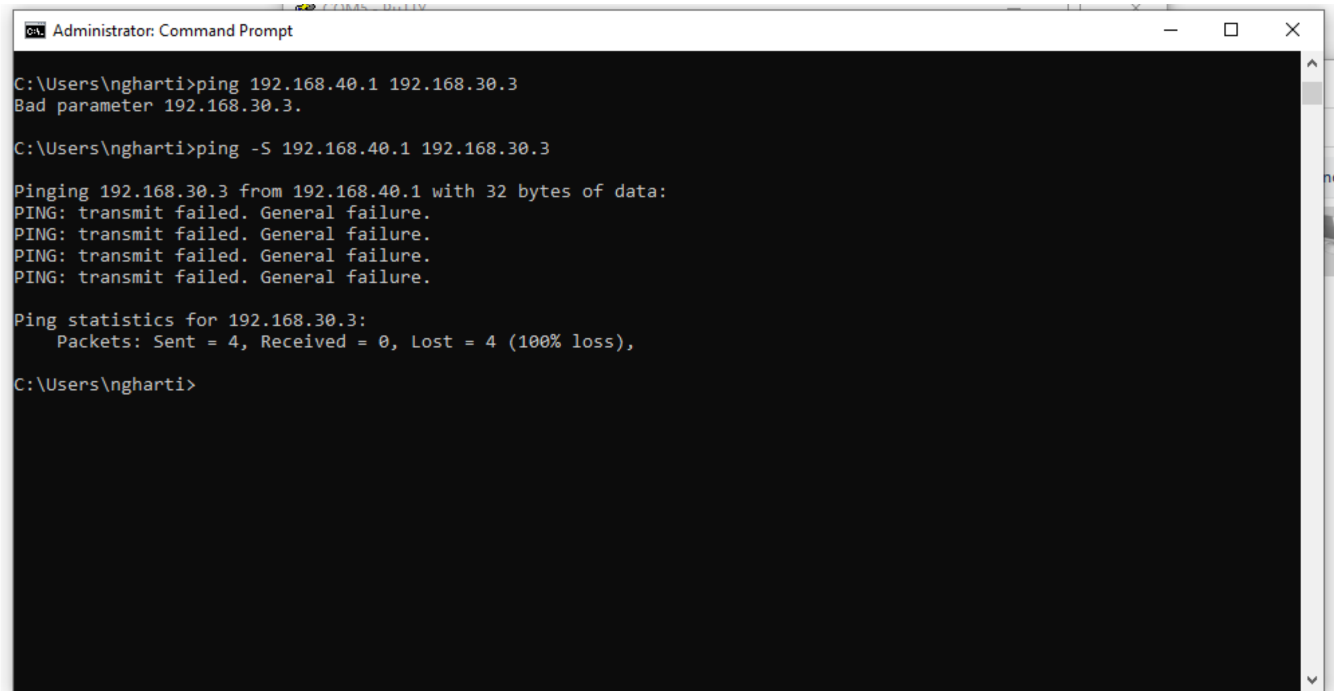



```
R3#show ip interface g0/1
GigabitEthernet0/1 is up, line protocol is up
  Internet address is 192.168.30.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Multicast reserved groups joined: 224.0.0.9
  Outgoing access list is 1
  Inbound  access list is not set
  Proxy ARP is enabled
  Local Proxy ARP is disabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is enabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP CEF switching is enabled
  IP CEF switching turbo vector
 --More--
```

3) Test the ACL to see if it allows traffic from the 192.168.10.0/24 network access to the 192.168.30.0/24 network. From the PC-A command prompt, ping the PC-C IP address. Were the pings successful? __**NO** _____

4) Test the ACL to see if it allows traffic from the 192.168.20.0/24 network access to the 192.168.30.0/24 network. You must do an extended ping and use the loopback 0 address on R1 as your source. Ping PC-C's IP address. Were the pings successful? ___**NO** ____

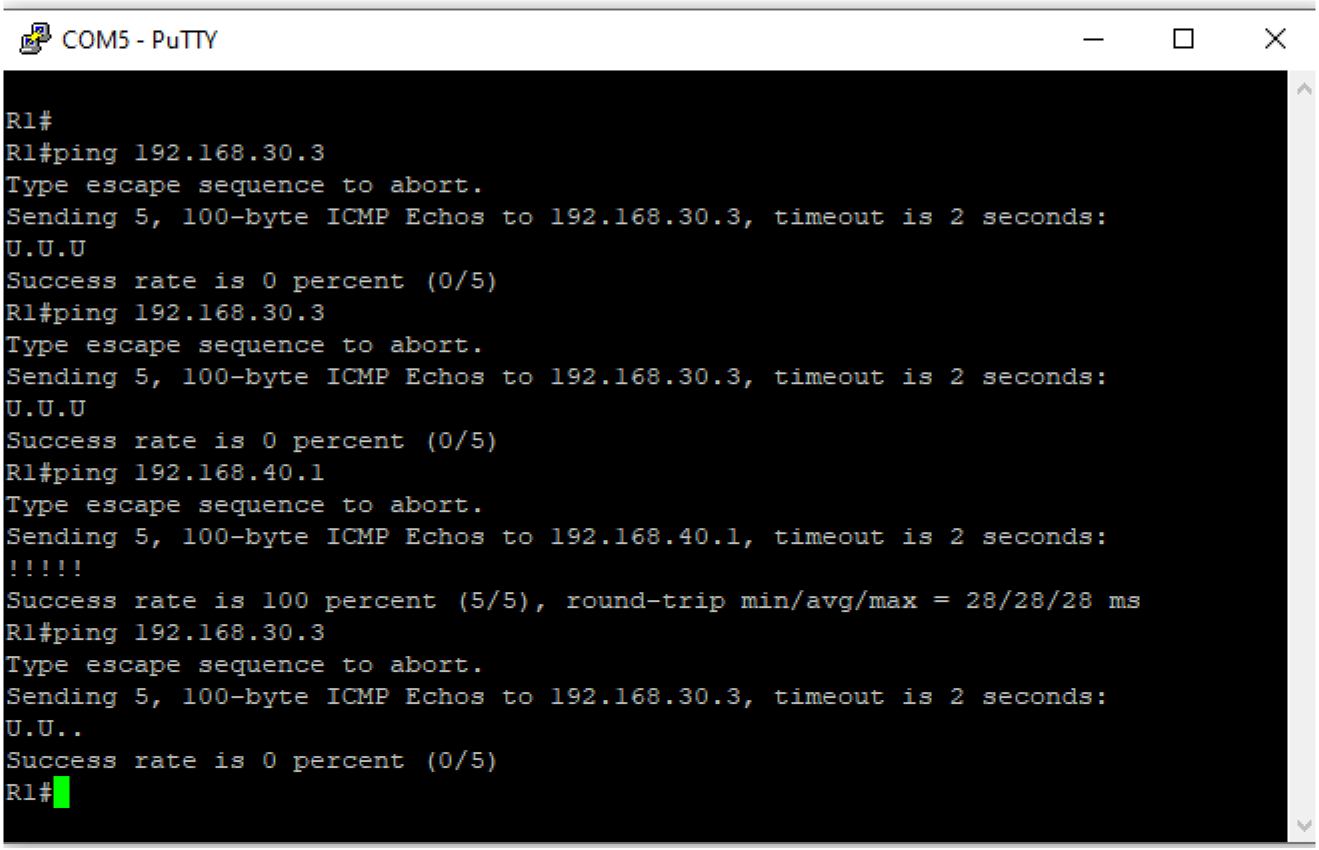
Insert here screen shots of successful or unsuccessful ping commands [1 mark]



d. From the R1 prompt, ping PC-C's IP address again.

R1# **ping 192.168.30.3**

Insert here screen shots of successful or unsuccessful ping commands **[1 mark]**



Was the ping successful? Why or why not?

Ping was not successful because there was network issues and configuration problems.

Step 2: Configure a named standard ACL.

Create a named standard ACL that conforms to the following policy: allow traffic from all hosts on the 192.168.40.0/24 network access to all hosts on the 192.168.10.0/24 network. Also, only allow host PC-C access to the 192.168.10.0/24 network. The name of this access list should be called BRANCH-OFFICE-POLICY.

Following Cisco's recommended best practices, on which router would you place this ACL? ___**R1**____

On which interface would you place this ACL? In what direction would you apply it?

I would apply the ACL "BRANCH-OFFICE-POLICY" on the outbound (out) direction of the interface GigabitEthernet0/1(g0/1) on Router R1.

a. Create the standard named ACL BRANCH-OFFICE-POLICY on R1.

```
R1(config)# ip access-list standard BRANCH-OFFICE-POLICY
R1(config-std-nacl)# permit host 192.168.30.3
R1(config-std-nacl)# permit 192.168.40.0 0.0.0.255
R1(config-std-nacl)# end
R1#
*Feb 15 15:56:55.707: %SYS-5-CONFIG_I: Configured from console by console
```

Looking at the first permit ACE in the access list, what is another way to write this?

____________**permit host 192.168.30.3**__

b. Apply the ACL to the appropriate interface in the proper direction.

```
R1# config t
R1(config)# interface g0/1
R1(config-if)# ip access-group BRANCH-OFFICE-POLICY out
```

c. Verify a named ACL.

1) On R1, issue the **show access-lists** command.

```
R1# show access-lists
Standard IP access list BRANCH-OFFICE-POLICY
    10 permit 192.168.30.3
    20 permit 192.168.40.0, wildcard bits 0.0.0.255
```

Is there any difference between this ACL on R1 with the ACL on R3? If so, what is it?

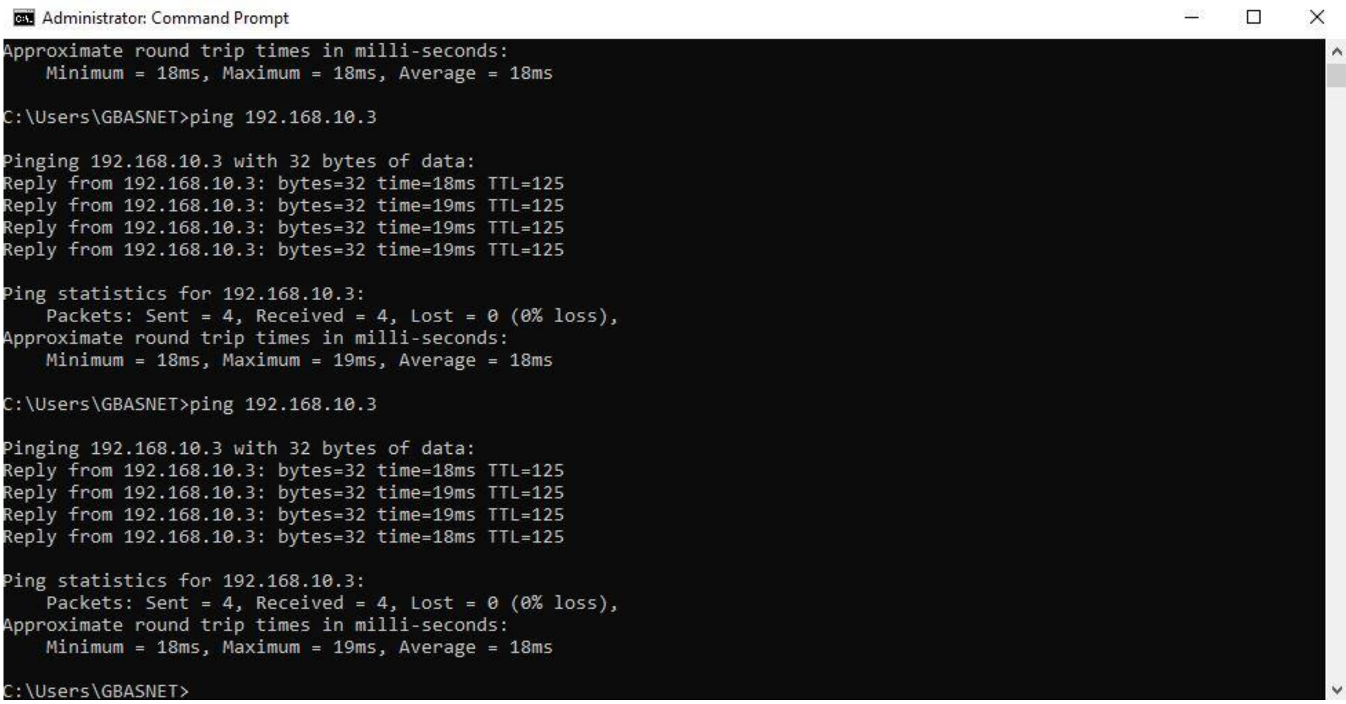
_______ **In conclusion, the primary distinction is that the 192.168.40.0/24 network is able to access the 192.168.10.0/24 network because of the ACL on R1** ____

On R1, issue the **show ip interface g0/1** command.

```
R1# show ip interface g0/1
GigabitEthernet0/1 is up, line protocol is up
  Internet address is 192.168.10.1/24
  Broadcast address is 255.255.255.255
  Address determined by non-volatile memory
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Multicast reserved groups joined: 224.0.0.10
  Outgoing access list is BRANCH-OFFICE-POLICY
  Inbound access list is not set
<Output omitted>
```

2) Test the ACL. From the command prompt on PC-C, ping PC-A's IP address. Were the pings successful? ____**YES**___

Insert here screen shots of successful or unsuccessful ping commands [1 mark]



```
Administrator: Command Prompt
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 18ms, Average = 18ms

C:\Users\GBASNET>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:
Reply from 192.168.10.3: bytes=32 time=18ms TTL=125
Reply from 192.168.10.3: bytes=32 time=19ms TTL=125
Reply from 192.168.10.3: bytes=32 time=19ms TTL=125
Reply from 192.168.10.3: bytes=32 time=19ms TTL=125

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 19ms, Average = 18ms

C:\Users\GBASNET>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:
Reply from 192.168.10.3: bytes=32 time=18ms TTL=125
Reply from 192.168.10.3: bytes=32 time=19ms TTL=125
Reply from 192.168.10.3: bytes=32 time=19ms TTL=125
Reply from 192.168.10.3: bytes=32 time=18ms TTL=125

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 19ms, Average = 18ms

C:\Users\GBASNET>
```

3) Test the ACL to ensure that only the PC-C host is allowed access to the 192.168.10.0/24 network. You must do an extended ping and use the G0/1 address on R3 as your source. Ping PC-A's IP address. Were the pings successful? ___**Yes**____

4) Test the ACL to see if it allows traffic from the 192.168.40.0/24 network access to the 192.168.10.0/24 network. You must perform an extended ping and use the loopback 0 address on R3 as your source. Ping PC-A's IP address. Were the pings successful? _**Yes**___

Part 4: Modify a Standard ACL

It is common in business for security policies to change. For this reason, ACLs may need to be modified. In Part 4, you will change one of the previous ACLs you configured to match a new management policy being put in place.

Management has decided that users from the 209.165.200.224/27 network should be allowed full access to the 192.168.10.0/24 network. Management also wants ACLs on all of their routers to follow consistent rules. A **deny any** ACE should be placed at the end of all ACLs. You must modify the BRANCH-OFFICE-POLICY ACL.

You will add two additional lines to this ACL. There are two ways you could do this:

OPTION 1: Issue a **no ip access-list standard BRANCH-OFFICE-POLICY** command in global configuration mode. This would effectively take the whole ACL out of the router. Depending upon the router IOS, one of the following scenarios would occur: all filtering of packets would be cancelled and all packets would be allowed through the router; or, because you did not take off the **ip access-group** command on the G0/1 interface, filtering is still in place. Regardless, when the ACL is gone, you could retype the whole ACL, or cut and paste it in from a text editor.

OPTION 2: You can modify ACLs in place by adding or deleting specific lines within the ACL itself. This can come in handy, especially with ACLs that have many lines of code. The retyping of the whole ACL or cutting and pasting can easily lead to errors. Modifying specific lines within the ACL is easily accomplished.

Note: For this lab, use Option 2.

Step 1: Modify a named standard ACL.

a. From R1 privileged EXEC mode, issue a **show access-lists** command.

```
R1# show access-lists
Standard IP access list BRANCH-OFFICE-POLICY
    10 permit 192.168.30.3 (8 matches)
    20 permit 192.168.40.0, wildcard bits 0.0.0.255 (5 matches)
```

b. Add two additional lines at the end of the ACL. From global config mode, modify the ACL, BRANCH-OFFICE-POLICY.

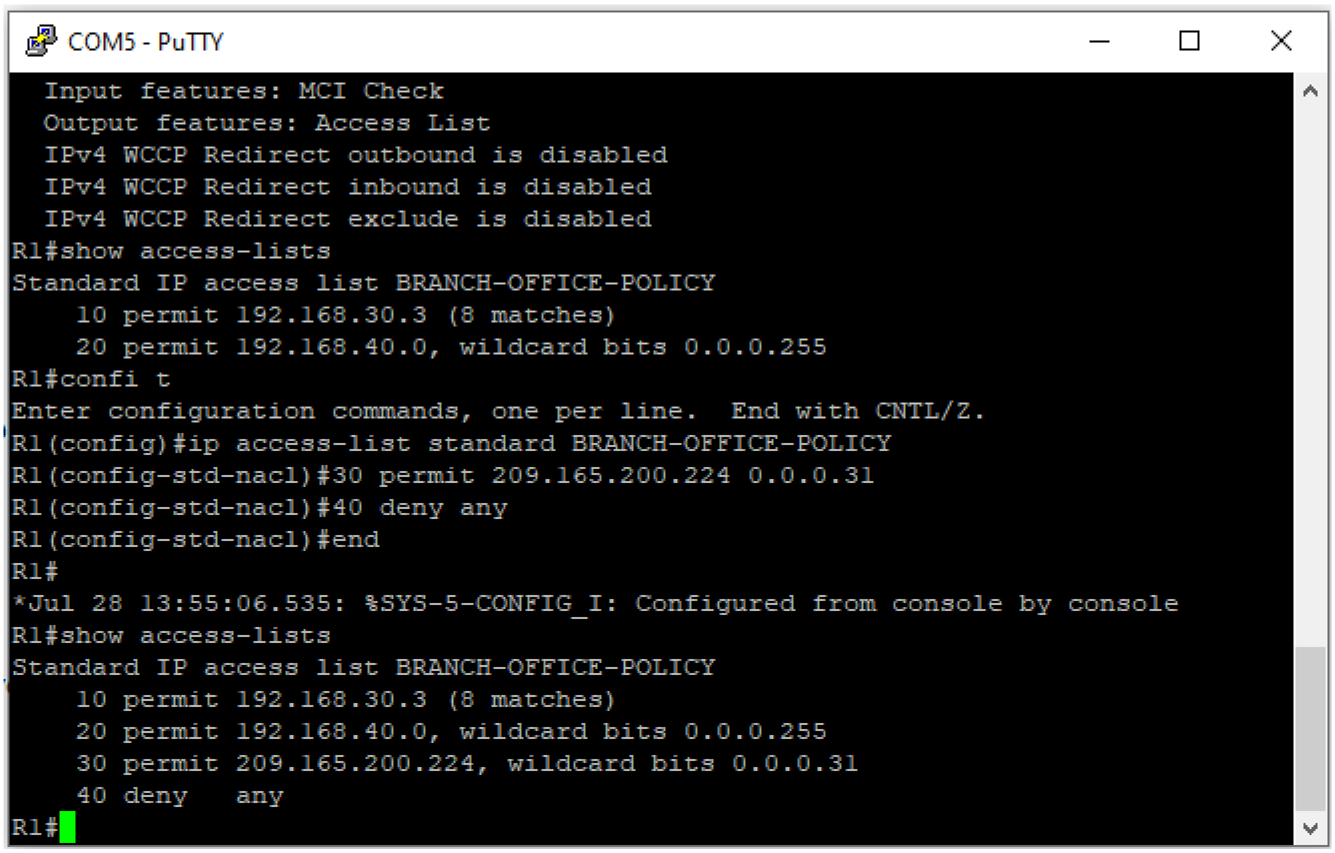
```
R1#(config)# ip access-list standard BRANCH-OFFICE-POLICY
R1(config-std-nacl)# 30 permit 209.165.200.224 0.0.0.31
R1(config-std-nacl)# 40 deny any
R1(config-std-nacl)# end
```

c. Verify the ACL.

1) On R1, issue the **show access-lists** command.

```
R1# show access-lists
```

Insert here screen shots of this command [1 mark]



```
COM5 - PuTTY
  Input features: MCI Check
  Output features: Access List
  IPv4 WCCP Redirect outbound is disabled
  IPv4 WCCP Redirect inbound is disabled
  IPv4 WCCP Redirect exclude is disabled
R1#show access-lists
Standard IP access list BRANCH-OFFICE-POLICY
    10 permit 192.168.30.3 (8 matches)
    20 permit 192.168.40.0, wildcard bits 0.0.0.255
R1#confi t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip access-list standard BRANCH-OFFICE-POLICY
R1(config-std-nacl)#30 permit 209.165.200.224 0.0.0.31
R1(config-std-nacl)#40 deny any
R1(config-std-nacl)#end
R1#
*Jul 28 13:55:06.535: %SYS-5-CONFIG_I: Configured from console by console
R1#show access-lists
Standard IP access list BRANCH-OFFICE-POLICY
    10 permit 192.168.30.3 (8 matches)
    20 permit 192.168.40.0, wildcard bits 0.0.0.255
    30 permit 209.165.200.224, wildcard bits 0.0.0.31
    40 deny   any
R1#
```

Do you have to apply the BRANCH-OFFICE-POLICY to the G0/1 interface on R1?

______YES, ACL "BRANCH-OFFICE-POLICY" must be applied to the Router R1's GigabitEthernet0/1 (G0/1) interface, according to the information given. On this interface, the ACL has to be applied to outgoing traffic. Accordingly, Router R1's GigabitEthernet0/1 interface's traffic will be filtered using the ACL. The ACL rules will be applied to all traffic leaving that particular interface as a result of this setup._____

2) From the ISP command prompt, issue an extended ping. Test the ACL to see if it allows traffic from the 209.165.200.224/27 network access to the 192.168.10.0/24 network. You must do an extended ping and use the loopback 0 address on ISP as your source. Ping PC-A's IP address. Were the pings successful? ___**YES**__

Reflection

1. As you can see, standard ACLs are very powerful and work quite well. Why would you ever have the need for using extended ACLs? **[1 mark]**

___ When filtering network traffic, extended ACLs offer greater granularity and flexibility. Extended ACLs provide you the ability to filter based on source, destination, particular protocols, and port numbers in addition to IP addresses, as opposed to conventional ACLs that only allow for source IP address-based filtering.__

Typically, more typing is required when using a named ACL as opposed to a numbered ACL. Why would you choose named ACLs over numbered? **[1 mark]**

___I would choose named ACL's over numbered to use meaningful name for access lists and ACE's which makes easier to understand. I would also use it for providing context for administrators which makes it clear what the ACL is intended for.__

Router Interface Summary Table

Router Interface Summary				
Router Model	**Ethernet Interface #1**	**Ethernet Interface #2**	**Serial Interface #1**	**Serial Interface #2**
1800	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
1900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2801	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/1/0 (S0/1/0)	Serial 0/1/1 (S0/1/1)
2811	Fast Ethernet 0/0 (F0/0)	Fast Ethernet 0/1 (F0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)
2900	Gigabit Ethernet 0/0 (G0/0)	Gigabit Ethernet 0/1 (G0/1)	Serial 0/0/0 (S0/0/0)	Serial 0/0/1 (S0/0/1)

Note: To find out how the router is configured, look at the interfaces to identify the type of router and how many interfaces the router has. There is no way to effectively list all the combinations of configurations for each router class. This table includes identifiers for the possible combinations of Ethernet and Serial interfaces in the device. The table does not include any other type of interface, even though a specific router may contain one. An example of this might be an ISDN BRI interface. The string in parenthesis is the legal abbreviation that can be used in Cisco IOS commands to represent the interface.